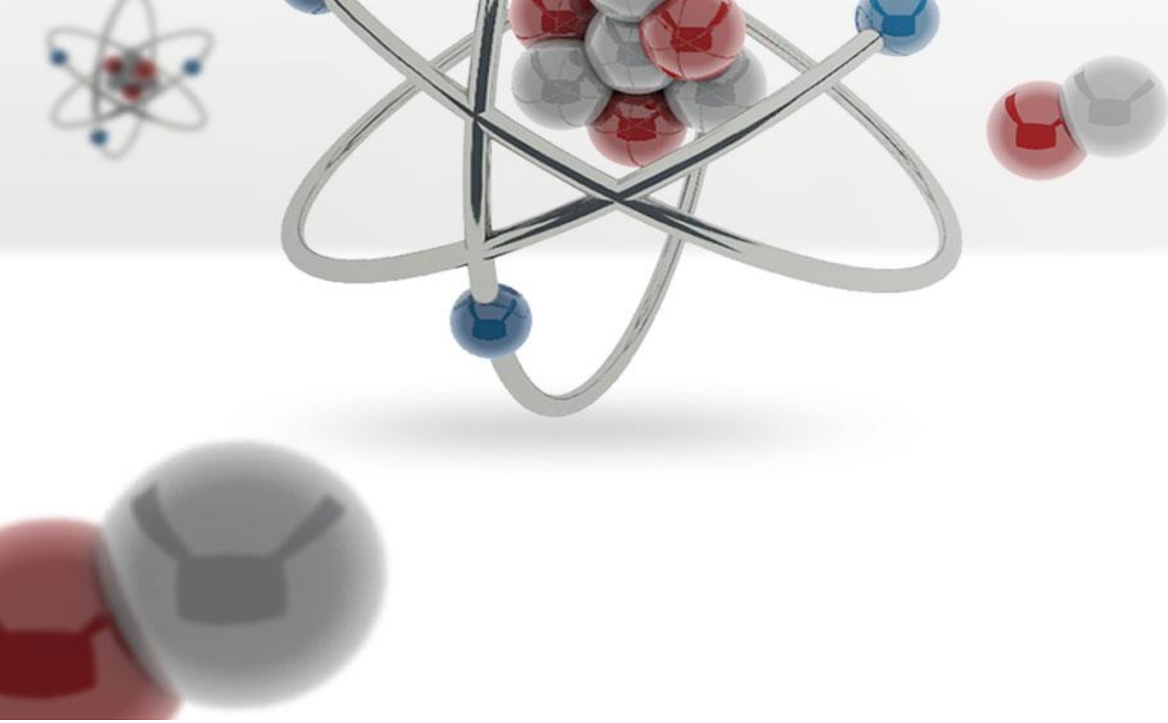

Computational Methods for Quantum Many-Body Systems (CMQMS) - from artificial atoms to high-temperature superconductors

Lecture 7 – Feynman diagrams and perturbation theory

UNIVERSITÀ DEGLI STUDI DI TRIESTE

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Università degli studi di Trieste (UNITS), Winter Semester 2025

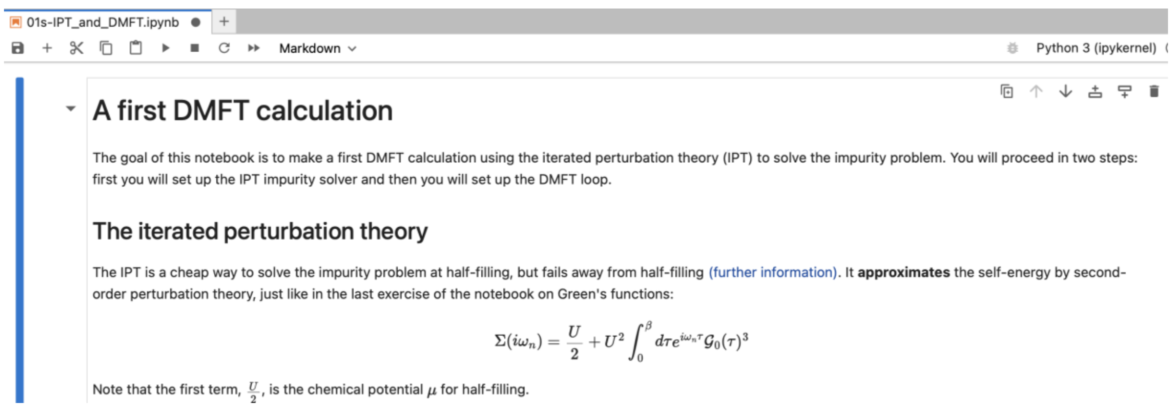
Tutorials

Please check, whether you can connect to:

http://sdsc-binder.flatironinstitute.org
Owner: ccq
Project: CMQMS

TRIQS

Toolbox for **R**esearch on **I**nteracting **Q**uantum **S**ystems


01s-IPT_and_DMFT.ipynb

Markdown

Python 3 (ipykernel)

A first DMFT calculation

The goal of this notebook is to make a first DMFT calculation using the iterated perturbation theory (IPT) to solve the impurity problem. You will proceed in two steps: first you will set up the IPT impurity solver and then you will set up the DMFT loop.

The iterated perturbation theory

The IPT is a cheap way to solve the impurity problem at half-filling, but fails away from half-filling (further information). It **approximates** the self-energy by second-order perturbation theory, just like in the last exercise of the notebook on Green's functions:

$$\Sigma(i\omega_n) = \frac{U}{2} + U^2 \int_0^\beta d\tau e^{i\omega_n \tau} \mathcal{G}_0(\tau)^3$$

Note that the first term, $\frac{U}{2}$, is the chemical potential μ for half-filling.

Please do not forget your laptops tomorrow!

Lecture 7

Content and goals

- Perturbation theory
- Feynman diagrams and their categorization
- Dyson equation and self-energy
- (non-)Fermi liquid form of the self-energy

Feynman diagram quiz